Validity Analysis: GSM8K

Target context: US Adult Personal Finance Chatbot Users

Overall risk: **HIGH**

Deployment Context

We are a US fintech startup building a personal finance chatbot that helps American adults with everyday money math: splitting bills, comparing loan APRs, calculating how long to pay off credit card balances, and budgeting across expense categories. Users type natural language questions about their finances. We want to evaluate whether an LLM can handle the quantitative reasoning these queries require before deploying the chatbot.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ■	1	Serious concern	high
Input Content ■	1	Serious concern	high
Input Form ✓	4	Minor gaps	high
Output Ontology ■	1	Serious concern	high
Output Content	2	Significant gaps	high
Output Form	3	Moderate gaps	high
Average	**2.0**		

■ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

GSM8K is a well-constructed grade-school math word-problem benchmark whose taxonomy, content, scoring ontology, and labels are explicitly scoped to elementary arithmetic in everyday consumer contexts. For a US adult personal-finance chatbot whose core value proposition is multi-step financial-instrument reasoning (APR, compound interest, amortization, payoff timelines) over assumption-sensitive queries with potentially range-valued correct answers, the misalignment is severe across the three HIGH-priority dimensions: Input Ontology (no financial-instrument problem types — 0/80 in dataset sample), Input Content (no consumer-finance vocabulary, no formula-selection or rate-disambiguation requirement), and Output Ontology (binary exact-match scoring incompatible with assumption-sensitive answer ranges). The Input Form dimension is well-aligned (typed English, CoT), and Output Content/Form are partially functional but cannot transfer to the deployment's financial-domain needs without substantial supplementation. The benchmark may still serve as a low-bar arithmetic-prerequisite filter and as infrastructure inspiration (calculator annotations, token-level verifier, socratic decomposition) for purpose-built financial evaluations.

Practical Guidance

What This Benchmark Measures

GSM8K measures a model's ability to perform multi-step natural-language arithmetic reasoning at grade-school complexity with traceable intermediate steps. For the deployment, this maps cleanly only to the commoditized simpler-tier queries (bill splitting, percentage budgeting) and to the prerequisite arithmetic skill underlying all financial computation. The strongest dimension is Input Form (text-in/text-out, English, CoT-required), which matches the deployment surface exactly.

Construct Depth

Shallow for the deployment's core value proposition. GSM8K provides zero direct evidence on compound interest, APR comprehension, amortization, payoff timelines, minimum-payment semantics, or credit utilization — the three HIGH-priority dimensions (IO, IC, OO) each scored 1. The benchmark also acknowledges its own complexity ceiling [Q48, Q101], and FinChain independently confirms that even frontier LLMs show clear limitations in symbolic financial reasoning that GSM8K does not surface [WEB-16]. A high GSM8K score is necessary but far from sufficient evidence of deployment readiness.

What Else You Need

(1) For IO/IC: adopt FinChain [WEB-16] to evaluate compound-interest and other symbolic financial-reasoning competence, supplemented with a bespoke evaluation covering consumer-specific query types (credit card payoff, minimum payment, balance transfer, credit utilization) since FinChain does not cover these. (2) For OO: implement assumption-pinned problem statements (FinChain-style parameterized templates) plus tolerance-band scoring for legitimately-range-valued queries. (3) For IC terminology grounding: build a targeted evaluation that tests term-to-operation mapping (APR→annualized periodic rate, minimum payment→issuer-formula-dependent) as a distinct construct from arithmetic — this gap is unaddressed by any benchmark surveyed [regional YAML gap 2]. (4) For OC: any new financial-content evaluation must use annotators with documented consumer-finance expertise, and the annotation framework must address the 1.7%-lower-bound labeling-error issue with greater rigor given regulatory exposure [WEB-5].

ID	Evidence
Q48	“Assuming a log-linear trend, we can naively extrapolate these results to estimate that a model with 10^16 parameters would be required to reach an 80% solve rate, when using the full GSM8K training set.” (p.6)
Q101	“We expect verification to scale well to problem distributions that require more complex mathematical reasoning, and we hope GSM8K supports the development of new methods that scale even better.” (p.12)
WEB-5	CFPB June 2023 chatbot spotlight identifies chatbots as UDAAP/Reg E/Reg Z compliance risk — making assumption disclosure and reasoning transparency operationally important (https://www.consumerfinance.gov/about-us/newsroom/cfpb-issue-spotlight-analyzes-artificial-intelligence-chatbots-in-banking/)
WEB-16	FinChain uses expert-curated parameterized symbolic templates with machine-verifiable Python traces — illustrates the level of annotation rigor required for financial content (https://arxiv.org/abs/2506.02515)

Input Ontology (IO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *GSM8K* | ***Context:*** *US Adult Personal Finance Chatbot Users*

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

GSM8K's taxonomy is explicitly scoped to grade-school math word problems [Q1, Q8, Q11], with the authors positioning it as a 'tractable goal' at the lower end of the difficulty spectrum [Q11, Q20]. The deployment team identified APR comparisons, compound interest, amortization, and payoff timelines as ~40-50% of expected queries and the product's core value proposition. Empirical sampling of 80 datapoints found zero compound-interest, amortization, APR, minimum-payment, credit-utilization, or balance-transfer problems; the single 'interest' problem [DATASET-D3] is explicitly labeled simple interest and applies a flat annual rate. The authors themselves acknowledge a complexity ceiling [Q48, Q49, Q50] and note hope that future work will scale to 'problem distributions that require more complex mathematical reasoning' [Q101], implicitly conceding GSM8K does not represent such distributions. For this HIGH-priority dimension, the taxonomy gap is total for the core-value query types.

ID	Evidence
Q1	"we introduce GSM8K, a dataset of 8.5K high quality linguistically diverse grade school math word problems." (p.1)
Q8	"We are releasing GSM8K, a dataset of 8.5K high quality problems at the grade school math level." (p.2)
Q11	"At the same time, GSM8K solutions depend only on elementary concepts, so achieving high test performance is a tractable goal." (p.2)
Q20	"The MATH dataset (Hendrycks et al., 2021) is larger and significantly more complex than GSM8K, but the high difficulty makes it challenging to accurately measure progress given the current capabilities of state-of-the-art language models." (p.4)
Q48	"Assuming a log-linear trend, we can naively extrapolate these results to estimate that a model with 10^16 parameters would be required to reach an 80% solve rate, when using the full GSM8K training set." (p.6)
Q49	"It is even harder to extrapolate along the data dimension, since performance does not appear to follow a log-linear trend." (p.6)
Q50	"Nevertheless, it appears likely that the 175B model would require at least two additional orders of magnitude of training data to reach an 80% solve rate." (p.6)
Q101	"We expect verification to scale well to problem distributions that require more complex mathematical reasoning, and we hope GSM8K supports the development of new methods that scale even better." (p.12)
D3	Ariella has $200 more in her son's saving account than Daniella has in her son's savings account. Ariella's account earns her simple interest at the rate of 10% per annum. If Daniella has $400, how much money will Arialla have after two years? — Simple interest problem — uses savings account framing but applies flat annual rate, not compound

Strengths

- Multi-step arithmetic reasoning with 3-6 sequential operations is exercised consistently [DATASET-D17, DATASET-D12], which is a prerequisite skill for any financial computation.

ID	Evidence
D12	There are enough provisions in a castle to feed 300 people for 90 days. After 30 days, 100 people leave the castle. How many more days are left until all the food runs out? — Rate/ratio problem — requires proportional reasoning across changing quantities
D17	Daniel has a collection of 346 video games. 80 of them, Daniel bought for $12 each. Of the rest, 50% were bought for $7. All others had a price of $3 each. How much did Daniel spend on all the games in his collection? — Multi-tier cost calculation with percentages — 3-step monetary reasoning

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories per deployment: APR comparisons, credit card payoff timelines, amortization schedules, compound interest, credit utilization, balance transfer cost/benefit, bill splitting, budgeting percentages. The first six are the core value proposition (~40-50% of queries).

IO-2: Yes — GSM8K omits all financial-instrument categories. The 80-example dataset sample contains zero compound interest, APR, amortization, credit card, minimum payment, balance transfer, or credit utilization problems. The lone interest example [DATASET-D3] is explicit simple interest. The paper scopes problems to 'elementary concepts' [Q11] and 'grade school math' [Q1, Q8].

IO-3: Most GSM8K categories (animals, food, school, household items, sports) are construct-irrelevant for a personal-finance chatbot. They are not harmful per se but consume evaluation weight without informing the deployment use case, per the dataset sample showing groceries, baseball cards, toy cars, cupcakes, etc. [DATASET-D1, DATASET-D14, DATASET-D22].

IO-4: Major content-validity gap: the dimension marked HIGH priority by the user has zero coverage for compound interest, APR, amortization, payoff timelines, and minimum-payment semantics. GSM8K's documented complexity ceiling [Q48, Q49, Q50] further means even harder GSM8K problems would not reach the structural complexity of multi-period amortization or recursive compounding.

ID	Evidence
Q1	“we introduce GSM8K, a dataset of 8.5K high quality linguistically diverse grade school math word problems.” (p.1)
Q8	“We are releasing GSM8K, a dataset of 8.5K high quality problems at the grade school math level.” (p.2)
Q11	“At the same time, GSM8K solutions depend only on elementary concepts, so achieving high test performance is a tractable goal.” (p.2)
Q48	“Assuming a log-linear trend, we can naively extrapolate these results to estimate that a model with 10^16 parameters would be required to reach an 80% solve rate, when using the full GSM8K training set.” (p.6)
Q49	“It is even harder to extrapolate along the data dimension, since performance does not appear to follow a log-linear trend.” (p.6)
Q50	“Nevertheless, it appears likely that the 175B model would require at least two additional orders of magnitude of training data to reach an 80% solve rate.” (p.6)
Q101	“We expect verification to scale well to problem distributions that require more complex mathematical reasoning, and we hope GSM8K supports the development of new methods that scale even better.” (p.12)
D1	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. On Wednesday Buddy buys 12 baseball cards. On Thursday he buys a third of what he had on Tuesday. — Multi-step counting problem with no financial domain content
D3	Ariella has $200 more in her son's saving account than Daniella has in her son's savings account. Ariella's account earns her simple interest at the rate of 10% per annum. If Daniella has $400, how much money will Arialla have after two years? — Simple interest problem — uses savings account framing but applies flat annual rate, not compound
D14	They bought 4 bags of Reese's for $9 per bag, 3 bags of Snickers for $5 per bag, and 5 bags of Skittles for $7 per bag. How much did the unicorn piñata and the treats cost altogether? — Grocery/party supply addition — simplest form of monetary arithmetic
D22	Anna baked 60 cupcakes. She gives away 4/5 of the cupcakes to her classmates. Of the remaining 1/5 of cupcakes, she eats 3 cupcakes. How many cupcakes does she have left? — Fraction subtraction — everyday context, zero financial domain content
WEB-16	FinChain uses expert-curated parameterized symbolic templates with machine-verifiable Python traces — illustrates the level of annotation rigor required for financial content (https://arxiv.org/abs/2506.02515)

Information Gaps

- Whether the test split (not sampled) contains any financial-instrument problems — unlikely given documented scope but not empirically ruled out.

Remediation

Gap 1: GSM8K's taxonomy contains no financial-instrument problem types covering the deployment's core value proposition (~40-50% of queries).

Recommendation: Pair GSM8K with FinChain [WEB-16] for compound-interest and symbolic-financial-reasoning coverage, and commission a bespoke benchmark of ~500-1000 consumer-finance scenarios (APR comparisons, credit card payoff timelines, amortization, minimum-payment semantics, credit utilization, balance transfer break-even) using parameterized templates that pin assumptions explicitly.

Gap 2: GSM8K is documented as having a complexity ceiling [Q48, Q49, Q50] that does not reach the structural complexity of multi-period amortization or recursive compounding even at its hardest.

Recommendation: Treat GSM8K performance as a necessary but non-sufficient prerequisite: require a minimum GSM8K threshold (e.g., 90%+) as a gate before deploying, but base deployment decisions on the financial-specific evaluations above. Do not extrapolate GSM8K headline numbers to financial reasoning competence.

ID	Evidence
Q48	“Assuming a log-linear trend, we can naively extrapolate these results to estimate that a model with 10^16 parameters would be required to reach an 80% solve rate, when using the full GSM8K training set.” (p.6)
Q49	“It is even harder to extrapolate along the data dimension, since performance does not appear to follow a log-linear trend.” (p.6)
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WEB-16	FinChain uses expert-curated parameterized symbolic templates with machine-verifiable Python traces — illustrates the level of annotation rigor required for financial content (https://arxiv.org/abs/2506.02515)

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *GSM8K* | ***Context:*** *US Adult Personal Finance Chatbot Users*

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

GSM8K's content was authored by Upwork/Surge AI contractors with no documented financial expertise [Q103, Q104], guided by GPT-3 seed questions [Q110, Q111] and a linguistic-diversity-within-elementary-concepts principle [Q9]. No annotator demographics — geographic, educational, financial-literacy — are reported. Empirically, the 80-example sample contains zero use of consumer-finance vocabulary (APR, minimum payment, balance transfer, credit utilization, amortization); even monetary problems use plain consumer-purchase framing [DATASET-D14, DATASET-D16, DATASET-D5]. The deployment team explicitly identified terminology-to-operation mapping as a distinct construct and a serious failure mode independent of arithmetic correctness. For this HIGH-priority dimension, GSM8K provides no scaffolding for the financial-vocabulary disambiguation the deployment requires.

ID	Evidence
Q9	“We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts.” (p.2)
Q103	“We initially collected a starting set of a thousand problems and natural language solutions by hiring freelance contractors on Upwork (upwork.com).” (p.15)
Q104	“We then worked with Surge AI (surgehq.ai), an NLP data labeling platform, to scale up our data collection.” (p.15)
Q110	“To assist contractors with writing questions, we provided seed questions automatically generated from a few-shot prompted 175B GPT-3 model.” (p.15)
Q111	“Contractors were allowed to use those seed questions directly, to use them as inspiration and make modifications, or to come up with their own questions entirely.” (p.15)
D5	Last year Jessica paid $1000 for rent, $200 for food, and $100 for car insurance each month. This year her rent goes up by 30%, food costs increase by 50%, and the cost of her car insurance triples because she was at fault in an accident. — Personal budgeting scenario — percentage increases on fixed monthly expenses
D14	They bought 4 bags of Reese's for $9 per bag, 3 bags of Snickers for $5 per bag, and 5 bags of Skittles for $7 per bag. How much did the unicorn piñata and the treats cost altogether? — Grocery/party supply addition — simplest form of monetary arithmetic
D16	Austin bought his seven friends each a robot. Each robot costs $8.75. He was charged $7.22 total for tax. He left with $11.53 in change. How much did Austin start with? — Retail transaction with tax and change — closest analog to everyday consumer arithmetic with dollar amounts

Strengths

- Monetary arithmetic with dollar amounts, percentages, and consumer-purchase framing is present in roughly 40-50% of sampled problems [DATASET-D16, DATASET-D5, DATASET-D6], providing some signal on US-style dollar/percentage handling — a prerequisite for the deployment's simpler bill-splitting and budgeting tier.
- Linguistic diversity is an explicit design principle [Q9] enforced via pairwise similarity scoring [Q113], reducing template-matching shortcuts.

ID	Evidence
Q9	“We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts.” (p.2)
Q113	“To ensure contractors were not re-using problem templates, we computed pairwise similarity scores between problems and used this to provide feedback to contractors.” (p.15)

D5	Last year Jessica paid $1000 for rent, $200 for food, and $100 for car insurance each month. This year her rent goes up by 30%, food costs increase by 50%, and the cost of her car insurance triples because she was at fault in an accident. — Personal budgeting scenario — percentage increases on fixed monthly expenses
D6	James decides to replace his car. He sold his $20,000 car for 80% of its value and then was able to haggle to buy a $30,000 sticker price car for 90% of its value. How much was he out of pocket? — Car resale/purchase — percentage of face value, single-step arithmetic
D16	Austin bought his seven friends each a robot. Each robot costs $8.75. He was charged $7.22 total for tax. He left with $11.53 in change. How much did Austin start with? — Retail transaction with tax and change — closest analog to everyday consumer arithmetic with dollar amounts

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — the deployment requires recognition of US-specific financial vocabulary (APR per TILA/Reg Z [WEB-7], minimum payment, credit utilization, balance transfer) embedded in colloquial natural-language queries. The financial register is explicitly central to the deployment per elicitation.

IC-2: Cultural framing of GSM8K (everyday US-style consumer life: food, animals, school) is broadly US-compatible per the regional YAML, and the dataset sample shows no obvious non-US monetary conventions; however, expert review of the source-culture flag remains deferred and unverified.

IC-3: INSUFFICIENT DOCUMENTATION — annotator demographics (geographic location, educational background, financial literacy, native language) are not reported anywhere in the paper. Contractors were sourced via Upwork and Surge AI [Q103, Q104] but no further information is provided.

IC-4: INSUFFICIENT DOCUMENTATION — no regional annotator pool composition is documented; impossible to assess representativeness against the US adult personal-finance-chatbot user population.

IC-5: Two concrete content gaps harm content validity: (a) complete absence of consumer-finance terminology across the 80-example sample [DATASET-D10, DATASET-D5, DATASET-D7]; (b) no problems requiring financial-formula selection or rate-type disambiguation — even the one explicit-interest problem pins the type as 'simple' [DATASET-D3, DATASET-D21].

ID	Evidence
Q103	“We initially collected a starting set of a thousand problems and natural language solutions by hiring freelance contractors on Upwork (upwork.com).” (p.15)
Q104	“We then worked with Surge AI (surgehq.ai), an NLP data labeling platform, to scale up our data collection.” (p.15)
D3	Ariella has $200 more in her son's saving account than Daniella has in her son's savings account. Ariella's account earns her simple interest at the rate of 10% per annum. If Daniella has $400, how much money will Arialla have after two years? — Simple interest problem — uses savings account framing but applies flat annual rate, not compound
D5	Last year Jessica paid $1000 for rent, $200 for food, and $100 for car insurance each month. This year her rent goes up by 30%, food costs increase by 50%, and the cost of her car insurance triples because she was at fault in an accident. — Personal budgeting scenario — percentage increases on fixed monthly expenses
D7	The price of a home is $98 per square foot (sq ft). The house is 2,400 sq ft and the barn out back is 1,000 sq ft. How much is this property? — Real estate price calculation — multiplication of unit price by area

D10	Ian won $100 in the lottery. He decided to use the money to pay off debts. He paid $20 to Colin. He then paid twice as much to Helen, as he had paid to Colin. Then finally, he paid half as much to Benedict, as he had paid to Helen. How much money, in dollars, does he have left after paying off debts? — Debt-payment problem — uses "debts" vocabulary but involves no interest, APR, or balance computations
D21	Ariella's account earns her simple interest at the rate of 10% per annum — Explicit "simple interest" label — benchmark treats this as equivalent in difficulty to non-financial problems, but distinguishes from compound
WEB-1	FINRA NFCS — only 27% of US adults answer ≥5/7 financial-knowledge questions; ~70% wrong on compound-interest doubling — confirms general-population annotators (the likely contractor pool composition) cannot reliably produce financial-domain labels (https://www.finrafoundation.org/sites/finrafoundation/files/2025-07/NFCS-Report-Sixth-Edition-July-2025.pdf)
WEB-7	CFPB Reg Z §1026.22 — Appendix J APR method pins one correct answer for closed-end loans but leaves credit-card effective-vs-nominal communication outside this resolution (https://www.consumerfinance.gov/rules-policy/regulations/1026/22/)

Information Gaps

- Annotator demographics, including geographic distribution, financial literacy, and educational background, are entirely undocumented.
- Source-culture flag from the benchmark metadata is unverified; expert sample review would help confirm US compatibility of problem framings.

Requires Expert Verification

- Whether any GSM8K problems reflect non-US monetary or institutional conventions (deferred in the regional YAML).

Remediation

Gap 1: No consumer-finance terminology in GSM8K; the term-to-operation mapping construct (e.g., APR→periodic rate) is untested by any surveyed benchmark.

Recommendation: Build a targeted terminology-grounding evaluation suite that varies surface phrasing of consumer-finance terms (APR vs. interest rate vs. finance charge; minimum payment vs. full balance) and tests whether the model selects the mathematically correct operation. Use TILA/Reg Z conventions [WEB-7] as the canonical ground truth where applicable, and document deviations from those conventions explicitly.

ID	Evidence
WEB-7	CFPB Reg Z §1026.22 — Appendix J APR method pins one correct answer for closed-end loans but leaves credit-card effective-vs-nominal communication outside this resolution (https://www.consumerfinance.gov/rules-policy/regulations/1026/22/)

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *GSM8K* | ***Context:*** *US Adult Personal Finance Chatbot Users*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Both benchmark and deployment are text-only English with standard Latin script [Q32, Q33], processed through GPT-family tokenizers [Q35, Q43]. The deployment's interaction modality is typed natural language, matching GSM8K's surface form. The chain-of-thought requirement [Q56, Q57] aligns with the 'show your work' expectation surfaced by CFPB regulatory scrutiny [WEB-5]. The socratic config additionally provides explicit sub-question scaffolding [DATASET-D20, DATASET-D15] that mirrors stepwise financial-calculation explanation. This LOWER-priority dimension is well-matched at the form level; minor concerns are pedagogical (additive vs. exponential growth framing [DATASET-D15]) rather than form-level.

ID	Evidence
Q32	“First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions.” (p.5)
Q33	“Moreover, this choice enables our models to develop verbal analytical skills and to produce solutions that are more readily interpretable by humans.” (p.5)
Q35	“For both methods, we use models from the GPT-3 family as our initialization, primarily focusing on the 175B and 6B model sizes.” (p.6)
Q43	“We perform finetuning by updating model parameters to minimize the cross-entropy loss over all training tokens.” (p.6)
Q56	“We also note that it is important to allow the model to generate the full natural language solution before outputting a final answer.” (p.7)
Q57	“If we instead finetune a 6B model to directly output the final answer without any intermediate steps, performance drops drastically from 20.6% to 5.2%.” (p.7)
D15	How many new cars will Bobby acquire in the first year? ** In the first year, Bobby will acquire 16 * .5 = <<16*.5=8>>8 new cars. ... How many cars will Bobby have after the third year? ** After the third year, he will have 36 + 18 = <<36+18=54>>54 cars in total. — Socratic format decomposes growth steps but each year is computed from prior year's total — additive not exponential framing
D20	Define a variable ** Let x represent the number of Chevys ... Combine like terms ** 11x+15=301 ... Divide by 11 ** x=<<26=26>>26 — Socratic config adds explicit algebraic reasoning sub-steps — useful for evaluating chain-of-thought decomposition
WEB-5	CFPB June 2023 chatbot spotlight identifies chatbots as UDAAP/Reg E/Reg Z compliance risk — making assumption disclosure and reasoning transparency operationally important (https://www.consumerfinance.gov/about-us/newsroom/cfpb-issue-spotlight-analyzes-artificial-intelligence-chatbots-in-banking/)

Strengths

- Text-only English input matches deployment modality exactly [Q32, Q33].
- Inline calculator annotations [Q40, DATASET-D14] provide a template for machine-verifiable arithmetic traces, useful infrastructure for a CFPB-relevant 'show-your-work' deployment.
- Socratic decomposition format [DATASET-D20] supports step-level reasoning evaluation, closer to what regulatory transparency requires.

ID	Evidence

Q32	“First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions.” (p.5)
Q33	“Moreover, this choice enables our models to develop verbal analytical skills and to produce solutions that are more readily interpretable by humans.” (p.5)
Q40	“To mitigate this issue, we train all models to use a calculator by injecting calculation annotations into the training set.” (p.6)
D14	They bought 4 bags of Reese's for $9 per bag, 3 bags of Snickers for $5 per bag, and 5 bags of Skittles for $7 per bag. How much did the unicorn piñata and the treats cost altogether? — Grocery/party supply addition — simplest form of monetary arithmetic
D20	Define a variable ** Let x represent the number of Chevys ... Combine like terms ** 11x+15=301 ... Divide by 11 ** x=<<26=26>>26 — Socratic config adds explicit algebraic reasoning sub-steps — useful for evaluating chain-of-thought decomposition

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions match: typed English text, ASCII + standard Unicode punctuation, numerals, decimal notation. Both benchmark and deployment use the same input encoding.

IF-2: US infrastructure supports typed text input; 90% smartphone ownership and 95% internet use [WEB-12, WEB-14] confirm the regional substrate.

IF-3: Domain-specific form note: deployment queries embed consumer-finance vocabulary in natural language, which is a content rather than form difference; surface form (typed English) is unchanged.

IF-4: No material form mismatches identified. Minor: the calculator-annotation format [Q40, Q119] is a benchmark-internal training artifact and is unlikely to appear in user queries, but this does not harm external validity since it is a training-time intervention not part of the input contract.

ID	Evidence
Q32	“First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions.” (p.5)
Q33	“Moreover, this choice enables our models to develop verbal analytical skills and to produce solutions that are more readily interpretable by humans.” (p.5)
Q40	“To mitigate this issue, we train all models to use a calculator by injecting calculation annotations into the training set.” (p.6)
Q56	“We also note that it is important to allow the model to generate the full natural language solution before outputting a final answer.” (p.7)
Q57	“If we instead finetune a 6B model to directly output the final answer without any intermediate steps, performance drops drastically from 20.6% to 5.2%.” (p.7)
Q119	“During training, there is no special distinction between the annotated tokens and the rest of the solution: they are all just tokens.” (p.17)
WEB-12	~95% of US adults use the internet; 90% own smartphone — typed-text deployment infrastructure is well-supported (https://www.pewresearch.org/short-reads/2026/01/08/internet-use-smartphone-ownership-digital-divides-in-u-s/)
WEB-14	90% US smartphone ownership; 81% mobile bank access — typed-text output modality is appropriate (https://www.pewresearch.org/internet/fact-sheet/mobile/)

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *GSM8K* | ***Context:*** *US Adult Personal Finance Chatbot Users*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

GSM8K's output ontology is binary exact-match against a single ground-truth final number [Q31, Q60], with the only background-knowledge concession being 'basic background knowledge, like the number of days in a week' [Q21]. No tolerance bands, no range-based correctness, no assumption-pinning, no partial credit. The deployment requires range-based or assumption-conditioned scoring because a meaningful share of target queries (payoff timelines, effective vs. nominal APR, budgeting recommendations) have legitimate answer ranges depending on compounding frequency, issuer-specific minimum-payment formulas, and timing conventions — confirmed by the user in Q3 of elicitation and by regulatory context [WEB-7] (Reg Z fixes APR conventions for closed-end loans but not for credit-card effective vs. nominal communications, and not for issuer-specific minimum-payment formulas). The 80-example sample confirms uniform single-answer #### N termination [DATASET-D9, DATASET-D4]. For this HIGH-priority dimension, the structural and content validity of the scoring scheme is fundamentally misaligned with the deployment.

ID	Evidence
Q3	“At test time, we generate many candidate solutions and select the one ranked highest by the verifier.” (p.1)
Q21	“Similar to CommonsenseQA, GSM8K includes questions that require basic background knowledge, like the number of days in a week.” (p.4)
Q31	“Verifiers are trained to judge the correctness of solutions, with the training signal determined solely by whether or not the solution reached the correct final answer.” (p.5)
Q60	“Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer.” (p.7)
D4	In the second year, she earns the same amount of interest, which $60 + $60 = $120 — Answer treats the second year interest as identical to year one — confirming this is simple interest, not compound
D9	Ten friends decide to get an end-of-year gift for their teacher. They plan to split the cost of the gift equally. But four of the group drop out. The remaining friends split the cost equally among themselves. If each share is now $8 more, how much does the gift cost, in dollars? — Algebraic bill-splitting — the closest GSM8K analog to the deployment's "bill splitting" use case
WEB-7	CFPB Reg Z §1026.22 — Appendix J APR method pins one correct answer for closed-end loans but leaves credit-card effective-vs-nominal communication outside this resolution (https://www.consumerfinance.gov/rules-policy/regulations/1026/22/)

Strengths

- For the well-defined deterministic problems GSM8K does cover, exact-match scoring is objectively verifiable and reproducible [Q60, DATASET-D9].

ID	Evidence
Q60	“Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer.” (p.7)
D9	Ten friends decide to get an end-of-year gift for their teacher. They plan to split the cost of the gift equally. But four of the group drop out. The remaining friends split the cost equally among themselves. If each share is now $8 more, how much does the gift cost, in dollars? — Algebraic bill-splitting — the closest GSM8K analog to the deployment's "bill splitting" use case

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output ontology is single category: exact numeric match to final answer [Q31, Q60]. For the deployment's assumption-sensitive queries (payoff timelines, effective APR, amortization total interest), this is too narrow.

OO-2: Missing categories specific to the deployment: (a) range/tolerance-band correctness for assumption-sensitive queries, (b) assumption-conditioned correctness (e.g., 'correct given monthly compounding'), (c) partial credit for correct reasoning with wrong arithmetic vs. correct arithmetic on wrong formula — the verifier visualization appendix [Q61, Q155] documents these as known failure modes but does not score them differently.

OO-3: The single-answer convention encodes an assumption that mathematical problems have one correct answer. This holds for grade-school arithmetic but does not generalize to consumer finance where compounding-frequency conventions, issuer-specific minimum-payment formulas [regional YAML], and timing assumptions create legitimate answer ranges.

OO-4: Significant misalignment exists; stakeholder-driven redesign is warranted. FinChain's CHAINEVAL framework [WEB-16] illustrates how parameterized symbolic templates can pin assumptions explicitly while evaluating both final-answer and step-level correctness.

OO-5: Both structural validity (the scoring construct misrepresents the answer-space structure of financial queries) and content validity (entire output categories like 'valid range' and 'assumption-conditioned correct' are missing) are harmed. The deployment's hardest queries cannot be scored meaningfully under this ontology.

ID	Evidence
Q21	“Similar to CommonsenseQA, GSM8K includes questions that require basic background knowledge, like the number of days in a week.” (p.4)
Q31	“Verifiers are trained to judge the correctness of solutions, with the training signal determined solely by whether or not the solution reached the correct final answer.” (p.5)
Q60	“Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer.” (p.7)
Q61	“In practice, some solutions will reach the correct final answer using flawed reasoning, leading to false positives.” (p.7)
Q155	“The final row contains a false positive, where the model makes a mistake on the second step, where it subtracts 400 from the price of a diamond jewel instead of a gold one. Verifiers occasionally make mistakes with performing this variable binding of quantities to their relationships.” (p.22)
WEB-7	CFPB Reg Z §1026.22 — Appendix J APR method pins one correct answer for closed-end loans but leaves credit-card effective-vs-nominal communication outside this resolution (https://www.consumerfinance.gov/rules-policy/regulations/1026/22/)
WEB-16	FinChain uses expert-curated parameterized symbolic templates with machine-verifiable Python traces — illustrates the level of annotation rigor required for financial content (https://arxiv.org/abs/2506.02515)

Remediation

Gap 1: Binary exact-match scoring cannot accommodate range-valued or assumption-conditioned correct answers for ~half the deployment's query mix.

Recommendation: Adopt assumption-pinned problem statements (FinChain CHAINEVAL pattern [WEB-16]) as the primary mechanism, supplemented by tolerance-band acceptance for issuer-variable quantities (minimum-payment formulas) and explicit multi-answer ground truth where conventions diverge (effective vs. nominal APR). Score both final-answer correctness and step-level reasoning consistency.

ID	Evidence
WEB-16	FinChain uses expert-curated parameterized symbolic templates with machine-verifiable Python traces — illustrates the level of annotation rigor required for financial content (https://arxiv.org/abs/2506.02515)

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *GSM8K* | ***Context:*** *US Adult Personal Finance Chatbot Users*

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Label correctness within GSM8K's own scope is supported by a re-solve verification pass [Q105, Q106] with a documented 1.7% disagreement rate [Q107, Q108], acknowledged as a lower bound [Q109]. However, this quality control was performed by the same Upwork/Surge AI contractor pool with no documented domain expertise [Q103, Q104], and no annotator demographic information is reported. For the deployment, the more severe issue is that GSM8K labels for financial-adjacent problems may be conceptually correct grade-school answers but pedagogically misleading for finance: DATASET-D2 treats 50% annual growth additively (still hits answer for n=3 by coincidence in this case but mis-frames the operation); DATASET-D3/D4 explicitly uses simple interest where a real savings account would compound. The verifier failure-mode appendix [Q61, Q153, Q155] also documents that flawed reasoning with correct final answers and surface ambiguity create labeling errors — issues that would be amplified in financial content. The MODERATE-priority dimension is partially functional (labels are correct for grade-school content) but not fit for downstream financial-label transfer.

ID	Evidence
Q61	"In practice, some solutions will reach the correct final answer using flawed reasoning, leading to false positives." (p.7)
Q103	"We initially collected a starting set of a thousand problems and natural language solutions by hiring freelance contractors on Upwork (upwork.com)." (p.15)
Q104	"We then worked with Surge AI (surgehq.ai), an NLP data labeling platform, to scale up our data collection." (p.15)
Q105	"After collecting the full dataset, we asked workers to re-solve all problems, with no workers re-solving problems they originally wrote." (p.15)
Q106	"We checked whether their final answers agreed with the original solutions, and any problems that produced disagreements were either repaired or discarded." (p.15)
Q107	"We then performed another round of agreement checks on a smaller subset of problems, finding that 1.7% of problems still produce disagreements among contractors." (p.15)
Q108	"We estimate this to be the fraction of problems that contain breaking errors or ambiguities." (p.15)
Q109	"It is possible that a larger percentage of problems contain subtle errors." (p.15)
Q153	"The second row contains a problem where the solution is correct, but the verifier has rated it as incorrect. This is potentially due to the ambiguity between the "4 times" and the "4 potatoes" in the problem description." (p.22)
Q155	"The final row contains a false positive, where the model makes a mistake on the second step, where it subtracts 400 from the price of a diamond jewel instead of a gold one. Verifiers occasionally make mistakes with performing this variable binding of quantities to their relationships." (p.22)
D2	Bobby has 16 toy cars, and the number of cars he has increases by 50% every year. How many toy cars will Bobby have in three years? — Percentage growth applied to toy cars — not compound interest
D3	Ariella has $200 more in her son's saving account than Daniella has in her son's savings account. Ariella's account earns her simple interest at the rate of 10% per annum. If Daniella has $400, how much money will Arialla have after two years? — Simple interest problem — uses savings account framing but applies flat annual rate, not compound
D4	In the second year, she earns the same amount of interest, which $60 + $60 = $120 — Answer treats the second year interest as identical to year one — confirming this is simple interest, not compound

Strengths

• Documented re-solve verification pass with disagreement-driven repair/discard [Q105, Q106] establishes a non-trivial quality floor for the labels that do exist.

• 1.7% disagreement rate [Q107] is empirically measured rather than asserted, with the limitation explicitly acknowledged as a lower bound [Q108, Q109].

ID	Evidence
Q105	“After collecting the full dataset, we asked workers to re-solve all problems, with no workers re-solving problems they originally wrote.” (p.15)
Q106	“We checked whether their final answers agreed with the original solutions, and any problems that produced disagreements were either repaired or discarded.” (p.15)
Q107	“We then performed another round of agreement checks on a smaller subset of problems, finding that 1.7% of problems still produce disagreements among contractors.” (p.15)
Q108	“We estimate this to be the fraction of problems that contain breaking errors or ambiguities.” (p.15)
Q109	“It is possible that a larger percentage of problems contain subtle errors.” (p.15)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: For grade-school arithmetic, ground-truth labels are mathematically determined and stakeholder-independent. For deployment-relevant financial questions, GSM8K provides no labels — and the one financial-adjacent label (simple interest, DATASET-D4) reflects a pedagogical convention that diverges from real consumer-savings behavior.

OC-2: Disagreement between original annotators and regional financial-domain experts is likely for any borderline financial-adjacent problem, since contractors had no documented financial expertise [Q103, Q104]. Regional YAML notes ~70% of US adults answer the NFCS compound-interest question incorrectly [WEB-1], implying the contractor pool is unlikely to skew higher on this dimension.

OC-3: INSUFFICIENT DOCUMENTATION — no Datasheet or Data Statement; the paper reports only that contractors came from Upwork [Q103] and Surge AI [Q104] with no demographics, expertise, or geographic information.

OC-4: Re-annotation by a financial-expert regional pool would be needed for any extension into consumer finance; FinChain's expert-curated symbolic templates [WEB-16] offer a model for this.

OC-5: INSUFFICIENT DOCUMENTATION — aggregation method is described as disagreement-triggered repair/discard [Q106]; minority annotator perspectives are not surfaced or analyzed.

OC-6: Convergent validity is unverified for any financial-content extension because the annotator pool's financial expertise is undocumented. External validity is harmed because the simple-interest convention used in the lone interest-adjacent problem [DATASET-D21] does not generalize to deployment query expectations.

ID	Evidence
Q60	“Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer.” (p.7)
Q103	“We initially collected a starting set of a thousand problems and natural language solutions by hiring freelance contractors on Upwork (upwork.com).” (p.15)

Q104	“We then worked with Surge AI (surgehq.ai), an NLP data labeling platform, to scale up our data collection.” (p.15)
Q106	“We checked whether their final answers agreed with the original solutions, and any problems that produced disagreements were either repaired or discarded.” (p.15)
Q109	“It is possible that a larger percentage of problems contain subtle errors.” (p.15)
Q155	“The final row contains a false positive, where the model makes a mistake on the second step, where it subtracts 400 from the price of a diamond jewel instead of a gold one. Verifiers occasionally make mistakes with performing this variable binding of quantities to their relationships.” (p.22)
D4	In the second year, she earns the same amount of interest, which \$60 + \$60 = \$120 — Answer treats the second year interest as identical to year one — confirming this is simple interest, not compound
D21	Ariella's account earns her simple interest at the rate of 10% per annum — Explicit "simple interest" label — benchmark treats this as equivalent in difficulty to non-financial problems, but distinguishes from compound
WEB-1	FINRA NFCS — only 27% of US adults answer ≥5/7 financial-knowledge questions; ~70% wrong on compound-interest doubling — confirms general-population annotators (the likely contractor pool composition) cannot reliably produce financial-domain labels (https://www.finrafoundation.org/sites/finrafoundation/files/2025-07/NFCS-Report-Sixth-Edition-July-2025.pdf)
WEB-16	FinChain uses expert-curated parameterized symbolic templates with machine-verifiable Python traces — illustrates the level of annotation rigor required for financial content (https://arxiv.org/abs/2506.02515)

Information Gaps

- Annotator demographics and financial expertise levels are entirely undocumented.
- Aggregation method beyond binary disagreement-repair/discard is unspecified.

Requires Expert Verification

- Whether any GSM8K problems contain subtle labeling errors that would matter for a deployment-adjacent skill (the authors acknowledge subtle errors may exist beyond the 1.7% bound [Q109]).

Remediation

Gap 1: Annotator demographics and financial expertise are undocumented; 1.7% disagreement is an explicit lower bound on labeling errors [Q108, Q109], and the contractor pool is unlikely to have financial-domain expertise.

Recommendation: For any financial-content extension or supplementary benchmark, recruit annotators with demonstrated consumer-finance expertise (e.g., CFP credential, lending operations experience). Publish a Datasheet/Data Statement documenting annotator demographics, financial background, and inter-annotator agreement broken down by problem type.

ID	Evidence
Q108	“We estimate this to be the fraction of problems that contain breaking errors or ambiguities.” (p.15)
Q109	“It is possible that a larger percentage of problems contain subtle errors.” (p.15)

Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: *GSM8K* | ***Context:*** *US Adult Personal Finance Chatbot Users*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

GSM8K's output form is step-by-step natural-language solutions terminating in a single final number [Q56, Q57, Q29, Q46], which partially aligns with a typed-text chatbot deployment. The chain-of-thought requirement supports the deployment's need for transparent reasoning (relevant for CFPB-flagged UDAAP/Reg Z risk [WEB-5]), and the calculator-annotation format [Q40, Q119, DATASET-D14] provides a template for arithmetic verifiability. However, the deployment also requires the model to produce assumption disclosures, caveats, and range-based answers — outputs that GSM8K's form contract does not exercise or score. The token-level verifier interpretability [Q147-Q149] is a constructive form-level feature that could be ported to financial reasoning. This MODERATE-priority dimension is functional at the surface level (text-in/text-out, CoT) but does not exercise the assumption-disclosure or range-presentation aspects the deployment needs.

ID	Evidence
Q29	"At test time, we judge performance by autoregressively sampling a single low temperature solution and checking whether the final answer is correct." (p.5)
Q40	"To mitigate this issue, we train all models to use a calculator by injecting calculation annotations into the training set." (p.6)
Q46	"Test performance is determined by a single low temperature (T = 0) sample for each test problem." (p.6)
Q56	"We also note that it is important to allow the model to generate the full natural language solution before outputting a final answer." (p.7)
Q57	"If we instead finetune a 6B model to directly output the final answer without any intermediate steps, performance drops drastically from 20.6% to 5.2%." (p.7)
Q119	"During training, there is no special distinction between the annotated tokens and the rest of the solution: they are all just tokens." (p.17)
Q147	"One benefit of the token-level verifiers is that these models become immediately interpretable: we can visualize the predicted value for each token and better understand how the verifier makes decisions on judging samples." (p.21)
Q148	"Above we present a visualization of the predicted values for five different cherry-picked questions and model completions, verified by a 175B token-level verifier that was trained on the full training set." (p.21)
Q149	"In the visualization, the background color of the text corresponds to the verifier score for that token, where red is low value (predicted incorrect) and green" (p.21)
D14	They bought 4 bags of Reese's for $9 per bag, 3 bags of Snickers for $5 per bag, and 5 bags of Skittles for $7 per bag. How much did the unicorn piñata and the treats cost altogether? — Grocery/party supply addition — simplest form of monetary arithmetic
WEB-5	CFPB June 2023 chatbot spotlight identifies chatbots as UDAAP/Reg E/Reg Z compliance risk — making assumption disclosure and reasoning transparency operationally important (https://www.consumerfinance.gov/about-us/newsroom/cfpb-issue-spotlight-analyzes-artificial-intelligence-chatbots-in-banking/)

Strengths

- Natural-language step-by-step solutions [Q32, Q33] match the deployment's text-out modality and CFPB transparency expectations [WEB-5].
- Token-level verifier provides per-token interpretability [Q147, Q148, Q149], a diagnostic capability transferable to financial reasoning evaluation.

• Socratic config sub-question structure [DATASET-D20] models the kind of stepwise explanation users may need in personal-finance chatbot contexts.

ID	Evidence
Q32	“First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions.” (p.5)
Q33	“Moreover, this choice enables our models to develop verbal analytical skills and to produce solutions that are more readily interpretable by humans.” (p.5)
Q147	“One benefit of the token-level verifiers is that these models become immediately interpretable: we can visualize the predicted value for each token and better understand how the verifier makes decisions on judging samples.” (p.21)
Q148	“Above we present a visualization of the predicted values for five different cherry-picked questions and model completions, verified by a 175B token-level verifier that was trained on the full training set.” (p.21)
Q149	“In the visualization, the background color of the text corresponds to the verifier score for that token, where red is low value (predicted incorrect) and green” (p.21)
D20	Define a variable ** Let x represent the number of Chevys ... Combine like terms ** 11x+15=301 ... Divide by 11 ** x=<<26=26>>26 — Socratic config adds explicit algebraic reasoning sub-steps — useful for evaluating chain-of-thought decomposition
WEB-5	CFPB June 2023 chatbot spotlight identifies chatbots as UDAAP/Reg E/Reg Z compliance risk — making assumption disclosure and reasoning transparency operationally important (https://www.consumerfinance.gov/about-us/newsroom/cfpb-issue-spotlight-analyzes-artificial-intelligence-chatbots-in-banking/)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (typed English text with embedded step-by-step reasoning) matches deployment needs at the surface level [Q32, Q33]; however, the form does not include assumption-disclosure, range-presentation, or caveat-emission constructs that the deployment requires for assumption-sensitive queries.

OF-2: Not applicable — text-only chatbot deployment; no TTS requirement per regional YAML.

OF-3: Digital-literacy assumption is reasonable for the target population [regional YAML, WEB-14]; ~90% smartphone ownership and 95% internet use support typed-text accessibility for the deployment's intended user base, though the 5% gap is documented [WEB-12].

OF-4: Form mismatches harming external validity: (a) GSM8K does not score whether a solution states its assumptions, (b) does not score whether a solution offers a range when appropriate, (c) does not score whether a solution flags issuer-specific or convention-dependent variability. These are explicit deployment requirements that the output form does not exercise.

ID	Evidence
Q29	“At test time, we judge performance by autoregressively sampling a single low temperature solution and checking whether the final answer is correct.” (p.5)
Q32	“First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions.” (p.5)
Q33	“Moreover, this choice enables our models to develop verbal analytical skills and to produce solutions that are more readily interpretable by humans.” (p.5)

Q46	“Test performance is determined by a single low temperature (T = 0) sample for each test problem.” (p.6)
Q56	“We also note that it is important to allow the model to generate the full natural language solution before outputting a final answer.” (p.7)
WEB-5	CFPB June 2023 chatbot spotlight identifies chatbots as UDAAP/Reg E/Reg Z compliance risk — making assumption disclosure and reasoning transparency operationally important (https://www.consumerfinance.gov/about-us/newsroom/cfpb-issue-spotlight-analyzes-artificial-intelligence-chatbots-in-banking/)
WEB-12	~95% of US adults use the internet; 90% own smartphone — typed-text deployment infrastructure is well-supported (https://www.pewresearch.org/short-reads/2026/01/08/internet-use-smartphone-ownership-digital-divides-in-u-s/)
WEB-14	90% US smartphone ownership; 81% mobile bank access — typed-text output modality is appropriate (https://www.pewresearch.org/internet/fact-sheet/mobile/)

Information Gaps

- GSM8K does not document any output convention for stating assumptions or producing ranges; whether such outputs would be rejected by the scoring harness or simply ignored is not specified.

Remediation

Gap 1: Output form does not exercise or score assumption-disclosure, caveat-emission, or range-presentation — explicit deployment requirements and CFPB-relevant compliance behaviors [WEB-5].

Recommendation: Extend the evaluation rubric to score (a) presence and accuracy of stated assumptions for assumption-sensitive queries, (b) appropriate use of ranges where conventions vary, (c) explicit flags when issuer-specific variation makes a definitive answer impossible. The token-level verifier interpretability pattern [Q147-Q149] can be adapted as scoring infrastructure for these constructs.

ID	Evidence
Q147	“One benefit of the token-level verifiers is that these models become immediately interpretable: we can visualize the predicted value for each token and better understand how the verifier makes decisions on judging samples.” (p.21)
Q148	“Above we present a visualization of the predicted values for five different cherry-picked questions and model completions, verified by a 175B token-level verifier that was trained on the full training set.” (p.21)
Q149	“In the visualization, the background color of the text corresponds to the verifier score for that token, where red is low value (predicted incorrect) and green” (p.21)
WEB-5	CFPB June 2023 chatbot spotlight identifies chatbots as UDAAP/Reg E/Reg Z compliance risk — making assumption disclosure and reasoning transparency operationally important (https://www.consumerfinance.gov/about-us/newsroom/cfpb-issue-spotlight-analyzes-artificial-intelligence-chatbots-in-banking/)